

JUSTIN DAVIS

Principal Engineer

CAREER OBJECTIVE

Innovative and results-driven Principal Engineer and UI Architect with over 17 years of experience delivering scalable, user-centric digital solutions across healthcare, finance, retail, and media sectors. Proven expertise in full-stack development, modern frontend frameworks, authentication systems, and microservices architecture. Adept at leading cross-functional teams, driving digital transformation, and aligning technology with strategic business goals. Seeking to leverage deep technical acumen and leadership experience to architect impactful, future-ready platforms that enhance user experience, performance, and business value.

PROFESSIONAL EXPERIENCE

September 2019–Present | Castlight Health

Principal Engineer

About Castlight Health / apree health

Castlight Health, now part of **apree health**, is a digital healthcare company focused on transforming how employees and individuals navigate the U.S. healthcare system. Through a unified, data-driven platform, apree health integrates benefits navigation, advanced primary care (APC), clinical advocacy, and personalized digital tools to improve health outcomes, increase engagement, and reduce overall healthcare costs. Its solutions address critical challenges in U.S. healthcare by simplifying complex journeys, promoting preventive care, and enabling timely, informed decision-making for millions of users.

Role Summary & Key Contributions

As a Principal Engineer, I provided architectural leadership across multiple high-impact, customer-facing and internal platforms spanning healthcare navigation, public health infrastructure, identity and security modernization, and AI-driven user experiences.

- **Architectural Leadership & Platform Modernization:** Led the design and implementation of large-scale frontend and platform architectures, including migrating legacy monolithic applications to a scalable microfrontend architecture with Angular, enabling independent deployments, faster release cycles, and improved developer productivity.
- **Multi-Tenant Identity & Security Innovation:** Designed and implemented a **multi-tenant OpenID Connect (OIDC) authentication framework** supporting seamless integration with enterprise identity providers (Auth0, Okta, Google,

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g <https://scholar.google.com/citations?user=zHSpASEAAAAJ&hl=en>

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EDUCATION

May 2007

Calicut University,
Kerala, India

**B.Tech in Computer
Science Engineering**

RELEVANT SKILLS

Full-Stack Web Development

UI Architecture & Development

Authentication & Security
Architecture

Microservices & Microfrontend
Architecture

DevOps & CI/CD Pipelines

Cross-functional Team
Leadership

etc.), incorporating PKCE, biometric-optimized sessions, real-time observability, and secure session management across web and mobile platforms.

- **Conversational AI & Intelligent Healthcare Navigation:** Served as Principal Engineer for **PRISM**, a next-generation **multi-agent conversational AI platform** built on Gemini 2.5, LangGraph, and GCP. The system delivers personalized, multimodal healthcare guidance by integrating claims, benefits, eligibility, and provider data, while seamlessly escalating complex cases to human care guides.
- **Public Health & National-Scale Systems:** Led engineering efforts for **Project Curelight / Vaccine Finder**, a nationally deployed platform supporting real-time vaccine inventory reporting to the CDC in collaboration with Boston Children's Hospital. The system played a critical role in the U.S. COVID-19 response and established apree health as a trusted public-sector partner.
- **Rapid Pandemic Response Solutions:** Delivered **Working Well**, a HIPAA-compliant Return-to-Work platform enabling organizations to safely manage workforce health, regulatory compliance, and daily health assessments during the COVID-19 pandemic.
- **Cross-Functional Leadership & Mentorship:** Coordinated closely with product, clinical, security, and data teams to deliver complex initiatives end-to-end. Championed code quality, architectural rigor, and mentorship, helping teams adopt best practices in scalability, security, and modern frontend engineering.

Impact

My work at Castlight/apree health contributed to nationally visible healthcare platforms, AI-driven innovation in benefits navigation, improved public health infrastructure, and scalable architectures that continue to support millions of users while reducing operational cost and complexity for employers and health systems.

September 2017 – July 2019 | Solgenie Technologies

UI Architect

As a technology leader at Solgenie Technologies, I spearheaded digital transformation initiatives for global clients, architecting and delivering scalable, high-impact solutions that drove operational efficiency and business growth.

SAGE Portfolio Management System (BNY Mellon) - Lead Engineer

- Led the end-to-end development of SAGE, a next-generation web platform for portfolio managers overseeing 401k and retirement funds.
- Unified multiple legacy systems into a single, intuitive Angular-based application, enabling secure, real-time portfolio management and analytics accessible from any location.
- Architected RESTful APIs and engineered background services to handle high-volume financial calculations.
- Implemented SignalR for real-time UI updates, ensuring timely and accurate portfolio insights.

Performance Optimization & Scalability

Digital Transformation Strategy

Agile/Scrum Project Management

Javascript, Angular, React

Android, iOS, Ionic

Node.js, Java Spring boot, PHP

Docker, Kubernetes, Jenkins

Gitlab, Jira, Confluence

- Leveraged MongoDB for secure trade package archiving and SSIS for robust ETL processes, ensuring compliance, auditability, and seamless data integration.
- Established automated CI/CD pipelines and rigorous testing frameworks, accelerating feature delivery and enhancing code quality.
- Empowered investment teams with centralized visibility, streamlined workflows, and advanced decision support, significantly reducing operational risk and IT overhead.

NFL Sunday Ticket Web Modernization (DIRECTV) - UI Architect

- Led the migration of DIRECTV's NFL Sunday Ticket from Flash/Flex to a modern Angular 5, HTML5, and CSS stack, transforming the platform for millions of users.
- Architected a modular, scalable frontend and integrated advanced video DRM for secure, high-quality content delivery.
- Embedded robust authentication systems and optimized performance for 4K streaming and multi-device support.
- Established CI/CD pipelines and implemented automated testing to ensure rapid, reliable releases.
- Collaborated cross-functionally and mentored engineers, ensuring seamless migration and future-proofing the platform for innovation.
- The reimagined NFL Sunday Ticket expanded access, drove subscriber growth, and positioned DIRECTV as a leader in digital sports streaming.

Key Responsibilities Across Projects:

- Drove project planning, resource allocation, and stakeholder engagement to deliver projects on time and within budget.
- Managed and mentored cross-functional teams, fostering a culture of technical excellence and collaboration.
- Defined technical standards, ensured regulatory compliance, and maintained high code quality through code reviews and best practices.
- Translated business requirements into technical solutions, aligning technology strategy with client objectives.

August 2016 – September 2017 | Amazecodes Solutions

UI Architect | Technical Director

Project: Deluxe Corporation

Deluxe Corporation is a leading provider of personalized products and business services, including design, website development, print marketing, promotional items, and fraud protection for small businesses and financial institutions.

Project Details:

- Led the development of a comprehensive product suite, including an end-to-end Product Detail Page (PDP) featuring REST API integration.
- Developed a scalable email marketing solution using PHP and Smarty, enabling mass promotional campaigns with detailed email open statistics.
- Built custom API services for advanced image processing (format conversion, trimming, background removal) utilizing the PHP Imagick library.
- Implemented Google Analytics for enhanced user behavior tracking and data-driven product improvements.

Responsibilities:

- Created and maintained the Product Backlog; facilitated the breakdown of Epics, Themes, and Features into actionable user stories.
- Participated in Agile ceremonies: daily Scrums, Sprint Planning, Reviews, and Retrospectives.
- Conducted code reviews and monitored code quality using Git-based workflows.
- Engaged in full stack development using AngularJS, jQuery, Bootstrap, CSS3, HTML5, SASS, GULP, NodeJS modules, and PHP (Slim).
- Utilized tools such as Git, SVN, Jira, Confluence, Redmine, and Yoeman for streamlined development and project management.
- Evaluated team performance and devised plans for continuous improvement.

Project: SoulCafe

SoulCafe is a global relationship platform for Indians, focusing on fostering deep, meaningful connections based on personality, values, and life goals.

Responsibilities:

- Defined and executed the technological strategy, including CI/CD pipelines and microservices architecture.
- Coordinated cross-functional teams (UI/UX, App Development, Infra, SEO, Marketing) across onsite and offshore locations.
- Oversaw the implementation of build management processes using Yoeman for code validation, minification, and compliance.
- Managed the Product Backlog and collaborated on the breakdown of Epics, Themes, and Features into sprint-ready user stories.
- Orchestrated Agile ceremonies: daily Scrum, Sprint Planning, Reviews, and Retrospectives.
- Led continuous integration and delivery initiatives to streamline deployment.
- Scoped and evaluated new technologies and tools for innovation and development excellence.
- Provided input on design feasibility and project timelines; planned budgets, team structures, and talent acquisition.
- Involved in full stack development (AngularJS, jQuery, Bootstrap, CSS3, HTML5, SASS, GULP, NodeJS modules, PHP Slim).
- Maintained code quality through rigorous Git code review processes and build management with Yoeman.

Key Skills & Tools:

Agile/Scrum, Full Stack Development, REST APIs, Microservices, PHP (Slim), AngularJS, NodeJS, Git, SVN, Jira, Confluence, Redmine, Yoeman, Google Analytics, PHP Imagick, CI/CD, Team Leadership, Cross-functional Coordination

August 2014 – August 2016 | Target Corporation

Lead Interactive Specialist

Guest Profile Recommendation Engine

Target's Guest Profile Recommendation Engine is a proprietary, in-house system designed to deliver highly personalized product recommendations across Target's digital platforms, integrating both online and in-store guest data.

Project Details:

- Led UI engineering for the product detail page (PDP), Quick Info, and other key e-commerce touchpoints, ensuring seamless integration with the new recommendation engine.

- Developed reusable front-end components and dynamic templates using HTML5, CSS3, JavaScript, jQuery, and Backbone.js, optimizing the display of real-time, personalized recommendations.
- Created a custom jQuery framework to manage recommendations across multiple site pages and built application-level code for efficient API and web service integration (AJAX, JSON, XML).
- Collaborated with UX teams to review wireframes and guide the development of a lean, responsive UI layer using semantic HTML5, SASS, Bootstrap, and RESTful architecture.
- Ensured cross-browser compatibility, accessibility compliance, and adherence to W3C standards.
- Applied Agile/SCRUM methodologies, participated in sprint planning, reviews, and retrospectives, and mentored team members on best practices and problem-solving.
- Enhanced omnichannel synergy by integrating recommendations between digital and physical stores, contributing to improved conversion rates and customer satisfaction.

Product Image Zoom Optimization

Project Details:

- Conceptualized, developed, and led the implementation of a custom jQuery-based product image zoom plugin for Target.com, replacing the legacy Adobe Scene7 solution.
- Engineered a client-side zoom system that preloads high-resolution images, enabling all zoom and pan interactions within the browser and reducing server/API calls by over 90%.
- Conducted A/B testing, performance tuning, and full-scale rollout, ensuring a seamless, responsive user experience for millions of users.
- Authored technical documentation and coordinated cross-functional teams through development, testing, deployment, and post-launch optimization.

Key Responsibilities:

- Front-end development using HTML5, CSS3, JavaScript, jQuery, Backbone.js, and SASS.
- UI/UX review, accessibility compliance, and responsive design.
- API integration and dynamic template creation.
- Agile process participation and team leadership.
- Performance optimization, cross-browser bug fixing, and technical documentation.

Impact:

Drove significant cost savings, improved site performance, and delivered scalable, innovative solutions for Target's high-traffic e-commerce platform.

March 2013 – August 2014 | Capgemini

Senior Consultant

Initially joined Target Corporation as a consultant and later transitioned to a full-time role, contributing to all listed projects during both engagements.

August 2011 – March 2013 | TE Connectivity

Senior Software Engineer

Project Overview:

Prism FlexWave DAS is TE Connectivity's flagship distributed antenna system, designed to deliver high-capacity, scalable wireless coverage for large venues such as stadiums, airports, hospitals, campuses, and urban environments. The system features a modular architecture, multi-frequency and multi-operator support, fiber-based transport, and robust remote management, enabling seamless integration and management of 2G, 3G, and 4G/LTE wireless services in challenging coverage areas.

Responsibilities:

- Led the development of user interfaces to enable customers and administrators to configure and manage hardware components within the Prism FlexWave DAS product suite.
- Designed and implemented dynamic, interactive web applications using PHP, HTML, CSS, JavaScript, jQuery, AJAX, XML, and JSON, ensuring a responsive and intuitive user experience for configuring complex DAS hardware.
- Developed automation frameworks and authored automated test cases using Selenium RC and Java, significantly improving the reliability, efficiency, and coverage of system validation processes.
- Collaborated with cross-functional teams to integrate new features, support modular expansion, and ensure compatibility with multi-operator and multi-frequency requirements.
- Ensured all UI components adhered to best practices for usability, accessibility, and W3C compliance, supporting a diverse range of customer environments.
- Participated in requirements analysis, technical design reviews, and continuous process improvement to enhance product scalability and maintainability.
- Provided technical documentation and user guides for both internal teams and end-users, supporting streamlined deployments and troubleshooting.

Key Skills:

PHP, JavaScript, HTML, CSS, AJAX, jQuery, XML, JSON, Java, Selenium RC, Automation Frameworks, UI/UX Design, Agile Development, Cross-functional Collaboration

Impact:

- Delivered robust, user-friendly configuration tools that empowered customers to deploy and manage large-scale DAS installations efficiently.
- Enhanced system reliability and reduced manual testing effort through automation, contributing to the successful deployment of FlexWave Prism in high-profile venues worldwide, including stadiums and transportation hubs

March 2009 – July 2011 | Cosmonet Solutions

Systems Analyst

January 2008 – January 2009 | SkeinSoft

PHP developer

Professional Peer Review & Evaluation Activities

Recognized for international peer-review leadership, serving as a judge and evaluator of scholarly, technical, and industry contributions across global conferences, journals, and award programs.

- Peer Reviewer, IEEE International Conference on Data Science and Intelligent Network Computing (ICDSINC)

- Peer Reviewer, International Conference on Cyber Security and Artificial Intelligence (ICCSAI 2025)
- Peer Reviewer, International Conference on Communication, Computer and Information Technology (IC3IT 2025)
- Peer Reviewer, International Conference on Computing, Communication and Automation (ICCCA 2025)
- Peer Reviewer, International Conference on Recent Trends in Electrical, Electronics, and Computing Technologies (ICRTEECT 2025)
- Peer Reviewer, IEEE International Conference on Cyber-Enabled Distributed Computing and Knowledge Discovery (CyberC 2025)
- Peer Reviewer, IEEE Future Networks World Forum (FNWF 2025)
- Reviewer, IEEE Senior Member Applications, evaluating candidates for elevated professional status
- Judge, Globee Awards, serving as a volunteer evaluator across four award categories
- Chapter Reviewer, IGI Global Scientific Publishing – Book Submission System
- Active reviewer for upcoming 2026 international conferences, continuing ongoing peer-review service

Scholarly Publications & Professional Articles

Author of peer-reviewed journal articles and major professional publications addressing advanced software architecture, AI systems, cybersecurity, and scalable digital platforms.

Peer-Reviewed & Indexed Journals

- *ScienceDirect* — Scholarly article on advanced software and systems architecture
 - <https://www.sciencedirect.com/science/article/pii/S2666659625000381>
- *Journal of Computer Science and Technology Studies (JCSTS)*
 - <https://al-kindipublisher.com/index.php/jcsts/article/view/10132>
 - DOI: <https://doi.org/10.32996/jcsts.2025.7.115>
- *International Journal of Scientific Research (IJSR)*
 - <https://www.ijsr.net/getabstract.php?paperid=SR25621033541>
 - DOI: <https://dx.doi.org/10.21275/SR25621033541>
- *European Journal of Computer Science and Information Technology (EJCSIT)*
 - <https://ejournals.org/ejcsit/wp-content/uploads/sites/21/2025/07/AI-Dri-ven-Morning-Briefing.pdf>
 - DOI: <https://doi.org/10.37745/ejcsit.2013/vol13n497695>
- *Journal of Information Systems Engineering and Management (JISEM)*
 - <https://www.jisem-journal.com/index.php/journal/article/view/13076>
 - DOI: <https://doi.org/10.52783/jisem.v10i60s.13076>
- *SARC Council Journals*
 - *SJECS* — Machine Learning-Powered Personalization Engine for E-Commerce
 - DOI: <https://doi.org/10.5281/zenodo.15766853>
 - *SJMD* — Multi-Tenant OpenID Connect Integration for Secure and Scalable Authentication

Major Professional & Trade Media

- *DZone* — In-App Browsers in Mobile Applications
 - <https://dzone.com/articles/in-app-browsers-in-mobile-apps>

Professional Memberships & Fellowships

Elected to senior and fellow-grade professional memberships recognizing sustained contributions, technical excellence, and leadership within the engineering and scientific community.

- **Senior Member**, IEEE
- **Fellow**, Institution of Electronics and Telecommunication Engineers (IETE)
- **Fellow**, National Society of Professional Engineers (NSPE)
- **Royal Fellow**, International Organization of Advanced Skill Development (IOASD)
- **Fellow**, Raptor Hackathon
- **Fellow**, NIPES
- **Full Member**, Sigma Xi — The Scientific Research Honor Society
- **BCS Fellowship**, The Chartered Institute for IT

Honors & Awards

Recipient of nationally and internationally recognized awards acknowledging leadership, innovation, and exceptional contributions to technology, digital health, and large-scale systems.

- **Stevie Awards (Bronze)** — Leadership in National Vaccine Data Infrastructure
- **Titan Innovation Award (Gold)** — Innovation in Digital Health Technology
- **Noble Award (Gold)** — Outstanding Achievement in Digital Process Optimization
- **Global Recognition Award (GRA)**
- **Rockstar Award**, Target Corporation
- **Rocker Award**, Target Corporation
- **Impact Award**, Castlight Health

